



ABSTRACT

RNNs trained on sequential tasks often develop low-dimensional dynamical structures that organize network activity. We investigate the conditions under which these dynamical representations are preserved or overwritten. We train an RNN on a 1-bit flip-flop task in which the network learns bistable dynamics, then apply perturbations to the trained input weights to observe the reconstructed dynamics. In full-rank RNNs, original attractors are eliminated within the first training epoch, however further analysis suggests that the network subsequently recovers task performance by transitioning to slow-manifold dynamics, demonstrating that functional recovery can arise from distinct dynamical mechanisms. Find code at <https://github.com/Mosherof/When-to-Reuse/settings>

Introduction

RNNs trained on sequential tasks often develop structured internal dynamics that allow them to maintain information over time. Rather than using the full high-dimensional state space, the network's activity often organizes around simpler dynamical structures, like fixed points, encoding task-relevant states [1, 2, 3]. Through training, the weights shape this dynamical system. However, when the network continues learning, gradient updates can modify the weights in ways that disrupt previously learned dynamics [4]. This phenomenon is commonly referred to as catastrophic forgetting.

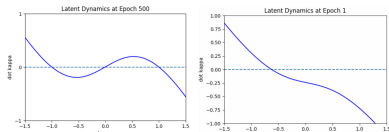


Fig 1. Catastrophic Forgetting after learning new task. A RNN was trained on a 1-bit flip-flop task and the latent dynamics revealed three fixed points. Initializing a new task with the learned parameters revealed a bifurcation by epoch 1.

Methods

Standard RNN Dynamics:

$$\tau \dot{\mathbf{r}}(t) = -\mathbf{r}(t) + \phi(\mathbf{W}\mathbf{r}(t) + \mathbf{W}^m\mathbf{u}(t) + \mathbf{b} + \epsilon)$$

$$\mathbf{r}[t+1] = (1-\alpha)\mathbf{r}[t] + \alpha \tanh(\mathbf{W}\mathbf{r}[t] + \mathbf{b} + \mathbf{W}^m\mathbf{u}[t])$$

We investigate both Rank-1 RNN and Full-Rank RNN on 1-bit flip flop task.

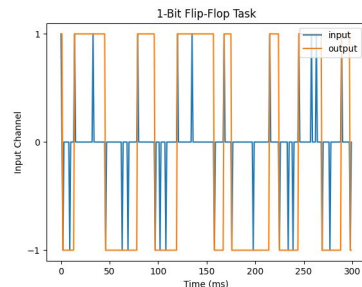


Fig 2. Example input and outputs of 1-bit flip-flop task.

Typically the 1-bit flip flop task is known to be solved through bistable attractor dynamics.

To investigate catastrophic forgetting we perturb the input and output weights in two different ways:

- Value Replacement: Replacing a fraction $(1-p)$ of weights with fresh samples from $N(0, 1/N)$.
- Reshuffling: Randomly permuting a fraction $(1-p)$ of existing entries, preserving the value distribution but scrambling the topology.

We compute fixed points and slow points through minimizing the kinetic energy function using mean square error:

$$\mathcal{K}(\mathbf{r}) = \frac{1}{2} \|\mathbf{F}(\mathbf{r})\|_2^2$$

Results

Rank-1 RNN Results:

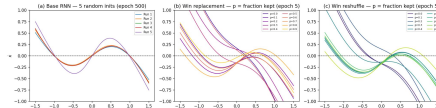
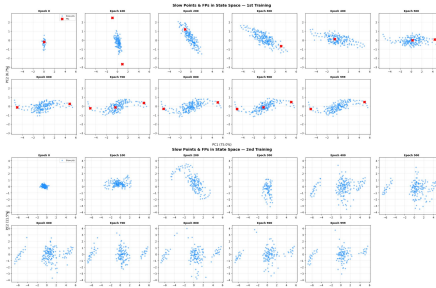


Figure 3. Summary of attractor dynamics under input weight perturbation after 5 epochs of retraining.

Reinitialized base networks all converge to the same bistable 3-fixed-point structure. Next, Low preservation of weights ($p \leq 0.4$) disrupts the attractor within 5 epochs, while high preservation leaves the 3-fixed-point structure largely intact. However, randomly reshuffling a fraction of Win entries is more robust against forgetting, confirming that out-of-distribution values rather than topology disruption drive catastrophic forgetting.

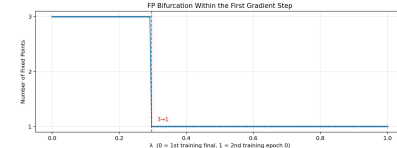
Full-Rank RNN Results:



weight perturbation

Even though the retrained network successfully recovered its task performance, the original bistable attractor landscape did not return. Instead, the network transitioned to an entirely different computational regime: a slow manifold.

Homotopy Continuation Test over first epoch after weight perturbation:



We show that the fixed points are being destroyed immediately after weight perturbation.

Conclusion

We challenge the common assumption that catastrophic forgetting is strictly a loss of functional capability.

Gradient descent can swap dynamical mechanisms entirely. Moving from discrete state-switching to continuous slow dynamics, while still maintaining or improving task performance.

Functional recovery does not necessitate structural restoration.

References

- [1] L. N. Driscoll, K. Shenoy, and D. Sussillo, "Flexible multitask computation in recurrent networks utilizes shared dynamical motifs", *en. Nature Neuroscience* 27, 1349–1363 (2024) 10.1038/s41593-024-01668-6.
- [2] B. Kurikaya, F. Dinc, E. Cirakman, M. Yuksekogul, M. Blanco-Pozo, E. Cirakman, M. Schnitler, Y. Yemez, H. Tanaka, P. Yuan, and N. Mialane, "Dynamical phases of short-term memory mechanisms in RNNs", *arXiv:2502.17433 [q-bio]*, June 2025, 10.48550/arXiv.2502.17433.
- [3] U. Haputhanthri, L. Storan, Y. Jiang, T. Raheja, A. Shai, O. Akenin, N. Mialane, M. J. Schnitler, F. Dinc, and H. Tanaka, "Understanding and controlling the geometry of memory organization in RNNs", *arXiv:2502.07256 [q-bio]*, Feb. 2025, 10.48550/arXiv.2502.07256.
- [4] D. Kunin, G. L. Marchetti, F. Chen, D. Karkada, J. B. Simon, M. R. DeWeese, S. Ganguli, and N. Mialane, "Alternating Gradient Flows: A Theory of Feature Learning in Two-layer Neural Networks", *arXiv:2506.06489 [cs]*, Dec. 2025, 10.48550/arXiv.2506.06489.
- [5] F. Dinc, E. Cirakman, Y. Jiang, M. Yuksekogul, M. J. Schnitler, and H. Tanaka, "A ghost mechanism: An analytical model of abrupt learning", *arXiv:2501.02378 [cs]*, Jan. 2025, 10.48550/arXiv.2501.02378.

Catastrophic Forgetting in RNN

Melissa Osherooff, Jasper Ou, Divyoj Singh, Justin Zhang

Introduction & Motivation

- RNN Standard Dynamics:

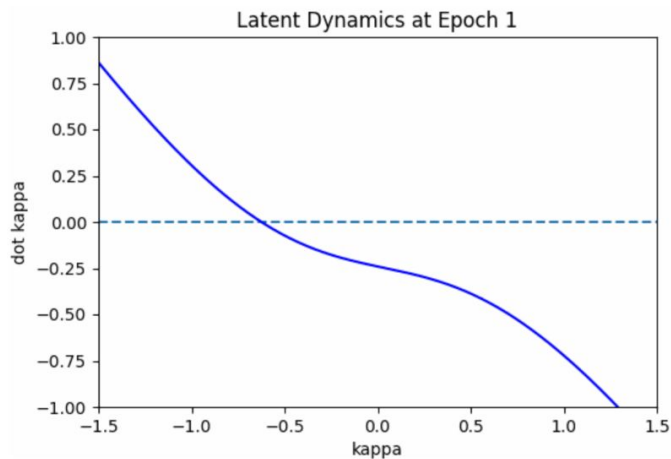
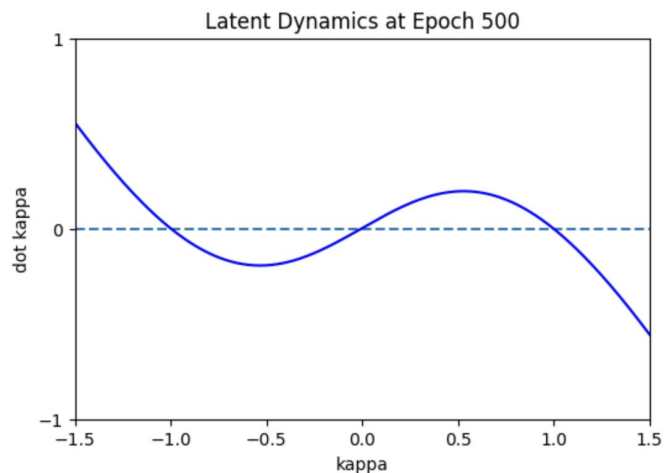
$$\tau \dot{\mathbf{r}}(t) = -\mathbf{r}(t) + \phi(\mathbf{W}\mathbf{r}(t) + \mathbf{W}^{in}\mathbf{u}(t) + \mathbf{b} + \epsilon)$$

$$r[t+1] = (1 - \alpha)r[t] + \alpha \tanh(\mathbf{W}r[t] + \mathbf{b} + \mathbf{W}^{in}u[t])$$

- As opposed to using the full high-dimensional state space, the network's activity often organizes around simpler dynamical structures
- Catastrophic Forgetting: learning a new sequential task rapidly disrupts or destroys previously acquired dynamical representations

Research Question

Under what conditions does gradient-based learning reuse existing computational structure, rather than overwriting it and constructing a new solution from scratch?



Methods

Investigated both Rank-1 RNN and Full-Rank RNN

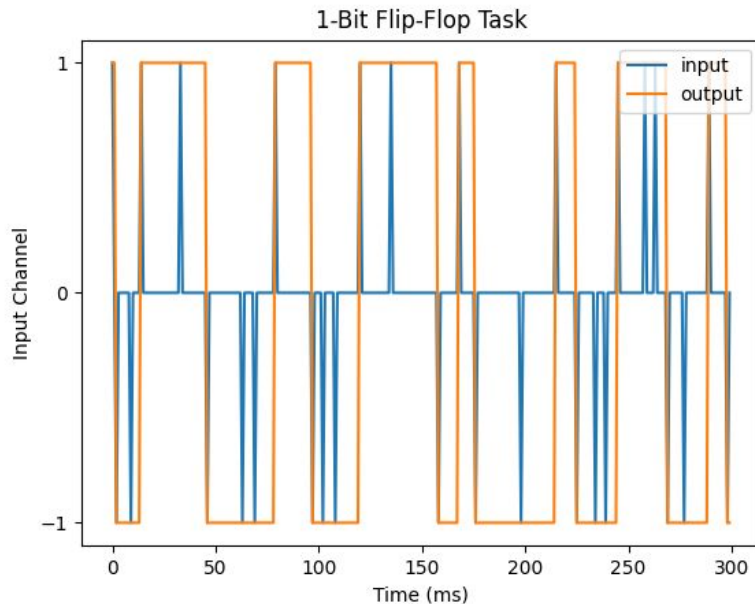
Task: 1-Bit Flip Flop

Finding fixed and slow points using kinetic energy minimization:

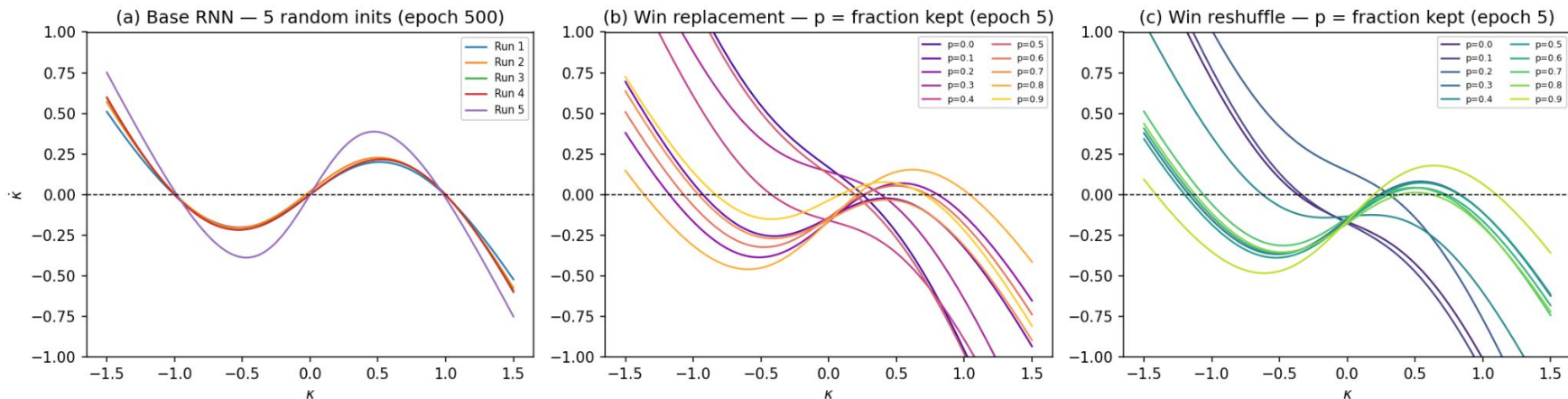
$$\mathcal{K}(\mathbf{r}) = \frac{1}{2} \|\mathbf{F}(\mathbf{r})\|_2^2$$

Perturbation Protocols:

1. Value Replacement: Replacing a fraction $(1 - p)$ of weights with fresh samples from $N(0, 1/N)$.
2. Reshuffling: Randomly permuting a fraction $(1 - p)$ of existing entries, preserving the value distribution but scrambling the topology.



Rank-1 RNN Results

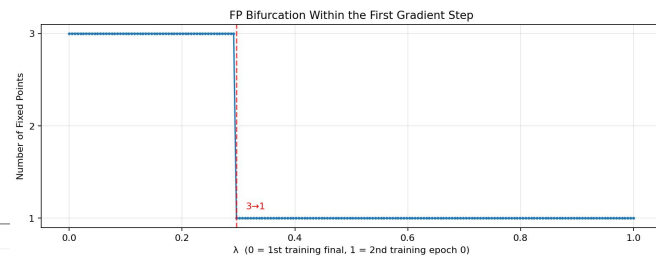


(a) Independently initialized base networks all converging to the same bistable 3-fixed-point structure

(b) Low preservation ($p \leq 0.4$) disrupts the attractor within 5 epochs, while high preservation ($p \geq 0.5$) leaves the 3-fixed-point structure largely intact during most runs

(c) Randomly permuting a fraction of Win entries is more robust to (b), confirming that out-of-distribution values rather than topology disruption drive catastrophic forgetting

Full-Rank RNN Results



Final Loss After 1st Training:
0.001



Final Loss After 2nd Training:
0.0008

Conclusion

We challenge the common assumption that catastrophic forgetting is strictly a loss of functional capability.

Gradient descent can swap dynamical mechanisms entirely. Moving from discrete state-switching to continuous slow dynamics, while still maintaining or improving task performance.

Functional recovery does not necessitate structural restoration.